

Reading CSV, Excel, JSON, HTML, and Pickle Files in Pandas

Pandas provides various functions to read and write data from different sources.

```
import pandas as pd
```

1. CSV File (Comma Separated Values)

CSV is the most commonly used file format for storing tabular data.

Sample CSV File

student.csv

```
id,name,age,city
1,Krish,20,Rajkot
2,Rahul,21,Ahmedabad
3,Priya,19,Surat
```

Read CSV

```
df = pd.read_csv("student.csv")
print(df)
```

Output

	id	name	age	city
0	1	Krish	20	Rajkot
1	2	Rahul	21	Ahmedabad
2	3	Priya	19	Surat

Common Parameters

Read Specific Columns

```
df = pd.read_csv("student.csv", usecols=["name", "age"])
```

Read First 2 Rows

```
df = pd.read_csv("student.csv", nrows=2)
```

Skip Rows

```
df = pd.read_csv("student.csv", skiprows=1)
```

Set Index Column

```
df = pd.read_csv("student.csv", index_col="id")
```

Handle Missing Values

```
df = pd.read_csv("student.csv", na_values=["NA", "NULL"])
```

2. Excel File

Excel files can contain multiple sheets.

Install Dependency

```
pip install openpyxl
```

Read Excel File

```
df = pd.read_excel("student.xlsx")  
print(df)
```

Read Specific Sheet

```
df = pd.read_excel("student.xlsx", sheet_name="Sheet1")
```

Read Multiple Sheets

```
data = pd.read_excel("student.xlsx", sheet_name=None)  
print(data.keys())
```

Output

```
dict_keys(['Sheet1', 'Sheet2'])
```

Write Excel File

```
df.to_excel("output.xlsx", index=False)
```

3. JSON File

JSON (JavaScript Object Notation) stores data as key-value pairs.

Sample JSON

student.json

```
[
  {
    "id":1,
    "name":"Krish",
    "age":20
  },
  {
    "id":2,
    "name":"Rahul",
    "age":21
  }
]
```

Read JSON

```
df = pd.read_json("student.json")
print(df)
```

Output

```
   id  name  age
0   1  Krish   20
1   2  Rahul   21
```

JSON String

```
json_data = '''
[
  {"id":1,"name":"Krish"},
  {"id":2,"name":"Rahul"}
]
'''
```

```
df = pd.read_json(json_data)
```

Write JSON

```
df.to_json("student.json", orient="records")
```

JSON Orient Types

Orient	Description
records	List of dictionaries
columns	Column-wise
index	Index-wise
values	Values only
table	Table schema

4. HTML File

Pandas can extract tables from HTML pages.

Sample HTML

```
<table>
<tr>
<th>ID</th>
<th>Name</th>
</tr>

<tr>
<td>1</td>
<td>Krish</td>
</tr>

<tr>
<td>2</td>
<td>Rahul</td>
</tr>
</table>
```

Read HTML Table

```
tables = pd.read_html("student.html")

print(tables[0])
```

Output

```
   ID  Name
0   1  Krish
1   2  Rahul
```

Read HTML From Website

```
tables = pd.read_html("https://example.com")

df = tables[0]
```

Important

`read_html()` always returns a list of DataFrames.

5. Pickle File

Pickle is used to save Python objects in binary format.

Very fast compared to CSV and Excel.

Extension:

```
.pkl  
.pickle
```

Create Pickle File

```
data = {  
    "Name": ["Krish", "Rahul"],  
    "Age": [20, 21]  
}  
  
df = pd.DataFrame(data)  
  
df.to_pickle("student.pkl")
```

Read Pickle File

```
df = pd.read_pickle("student.pkl")  
  
print(df)
```

Output

	Name	Age
0	Krish	20
1	Rahul	21

Advantages of Pickle

Faster

```
df.to_pickle("data.pkl")
```

Preserve Data Types

```
df["Date"] = pd.to_datetime(df["Date"])
```

CSV may lose exact datatype information, but Pickle preserves it.

Store Complex Objects

```
import pickle  
  
data = {  
    "name": "Krish",  
    "marks": [90, 80, 85]
```

```
}  
  
with open("data.pkl", "wb") as f:  
    pickle.dump(data, f)
```

Comparison Table

Feature	CSV	Excel	JSON	HTML	Pickle
Human Readable	Yes	Yes	Yes	Yes	No
Fast	Medium	Slow	Medium	Medium	Very Fast
Multiple Sheets	No	Yes	No	No	No
Stores Nested Data	No	No	Yes	No	Yes
Preserves Data Types	No	Partial	Yes	No	Yes
File Size	Small	Large	Medium	Medium	Small
Used For	Data Exchange	Reports	APIs	Web Tables	Python Storage

Most Used Reading Functions

```
pd.read_csv()  
pd.read_excel()  
pd.read_json()  
pd.read_html()  
pd.read_pickle()
```

Most Used Writing Functions

```
df.to_csv()  
df.to_excel()  
df.to_json()  
df.to_html()  
df.to_pickle()
```

Interview Questions

1. Why is Pickle faster than CSV?

Answer:

Pickle stores data in binary format, so reading and writing are much faster than text-based CSV files.

2. Which format is commonly used for APIs?

Answer:
JSON.

3. Which format supports multiple sheets?

Answer:
Excel (.xlsx).

4. Which function reads tables from webpages?

Answer:
`pd.read_html()`

5. Which format preserves Python object types?

Answer:
Pickle (.pkl).

Link : [CHATGPT](#)

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Read Multiple Sheets

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data = pd.read_excel("student.xlsx", sheet_name=None)
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```

Output

```
dict_keys(['Sheet1', 'Sheet2'])
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Write Excel File

```
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```

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df = pd.read_json("student.json")
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```

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JSON String

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</tr>
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df.to_pickle("student.pkl")
```

Read Pickle File

```
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Output

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Link : [CHATGPT](#)

Link : [ALL](#)